

Karamaniya: Measuring Persistent Multi-Agent AI Behavior

Two-page research brief for Anthropic AI for Science

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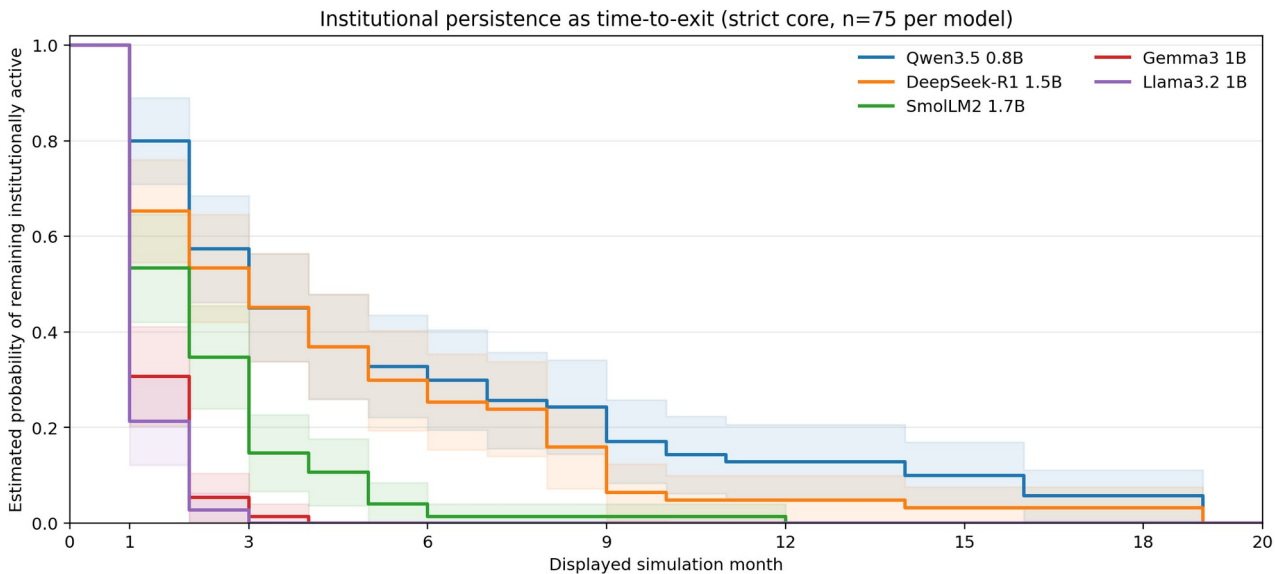
Pilot	Question	Request
85 completed trajectories 664 simulated months	Which long-horizon signatures survive corrected instrumentation?	\$30,000 staged Claude credit ceiling

Research objective

This is measurement science: building and validating an instrument for a new class of persistent multi-agent systems. Karamaniya places five language-model agents in recurring institutional roles inside one seeded causal world. Agents receive asymmetric evidence, communicate privately, bargain, vote, issue actions, accumulate memory, and experience consequences generated by a separate deterministic world engine. The goal is not to predict human politics; it is to measure persistent AI behavior under shared institutional constraints.

What the local pilot established

- The harness is operational end-to-end: raw and parsed model outputs, validation findings, votes, canonical state checkpoints, memory, outcomes, and report artifacts are recorded for each trajectory.
- A strict 75-run clean/full-formation core shows large differences in observed time to institutional exit under the current action interface. Qwen3.5-0.8B and DeepSeek-R1-1.5B remain active much longer than Gemma3-1B, Llama3.2-1B, and SmolLM2-1.7B, but this is treated as a candidate signal rather than an intrinsic model trait.
- The pilot surfaced its own confounds. State survival can overstate government functionality when offices are vacant, and resignation is a direct boolean action. A post-hoc audit found 100 of 328 canonical resignations had only placeholder-like notes/factors. These findings motivate instrument repair before frontier inference.
- Final unanimity is an exploratory secondary signal: 24 unanimous votes occurred, 19 of which the current automated classifier labels negotiated convergence. Human validation is required before confirmatory use.



Kaplan-Meier descriptive estimate; any recorded removal is an event. Members still active when a run terminates are censored.

Pilot strict-core Kaplan-Meier curves: descriptive all-cause institutional persistence under the pilot action interface. Curves validate discrimination, not governance quality.

Anthropic-supported study

Stage	Design	Purpose
A. Repair + bridge	Fix vacancy/resignation semantics; replay ~25 matched tiny-model seeds locally; randomize/reverse resignation-field order.	Measure instrument drift and test action-menu artifacts.
B. Calibration	Run ~10 Claude Sonnet 5.5 trajectories before locking the grid.	Measure full-government calls, token growth, cache utilization, and effort cost.
C. Main grid	~300 Sonnet baselines, ~150 matched Opus runs, ~300 Sonnet seat-swap/ablation runs.	Estimate model/effort, office, and mechanism effects.
D. Holdout	~150 untouched Sonnet holdout trajectories plus blinded human label validation.	Prospective replication of preregistered hypotheses.

Primary analysis

Primary persistence endpoint: time to institutional exit with explicit competing causes. Confirmatory analysis will use Kaplan-Meier all-cause curves, Aalen-Johansen cumulative incidence, and prespecified cause-specific Cox models with run-level dependence handled through robust clustering/frailty sensitivity analysis. The unanimity endpoint will be P(initial material disagreement | final unanimous vote) with frozen definitions and blinded human validation.

Transparent compute envelope

Component	Scale	Planning basis	Ceiling
Sonnet 5.5	760 trajectories	4x pilot input + conservative high-effort billed-output allowance	\$13.7k
Opus 5.5	150 trajectories	Same conservative token envelope	\$5.4k
Clean rerun/evaluator reserve	-	25% of model-call ceiling	\$4.8k
Calibration/variance reserve	-	20% after rerun reserve	\$4.8k
Calculated / requested	910 Claude trajectories	~\$28.6k calculated; rounded staged ceiling	\$30.0k

The raw token-only arithmetic for the same 910-run split is approximately \$5.3k-\$10.6k at 2-4x pilot long-run volume before caching. The \$30k figure is therefore a ceiling, not expected spend. The 10-run calibration is a hard budget gate; trajectory counts will be resized before confirmatory execution if observed costs differ materially. Prompt caching and batch savings are upside, not assumed in the base ceiling.

Why Anthropic compute changes the science

The local RTX 4060 pilot can establish lower-bound behavior and instrumentation, but it cannot answer whether candidate signatures survive at frontier capability and across Claude effort/model classes. Anthropic credits enable repeated long-horizon trajectories in which all five agents remain active, allowing role effects, resignation semantics, consensus dynamics, and ablations to be tested under a frozen harness rather than inferred from sparse or quota-truncated runs.

Eligibility and expected outputs

Anthropic's Official Rules state that an individual applicant must be at least 18 years old or, if greater, the age of majority in the applicant's jurisdiction; institutional consent applies when the applicant is employed by or affiliated with an institution [2]. Expected outputs are a versioned benchmark harness, corrected/validated trajectory dataset, preregistration, human-validated labels, and a public research paper on persistent multi-agent behavior.

Sources: [1] Anthropic, "Expanding our support for scientists," 27 Aug 2026. [2] Anthropic, "AI for Science Program Official Rules," updated 25 Aug 2026. [3] Anthropic, "Claude Sonnet 5.5," 28 Sep 2026. [4] Anthropic, "Claude Opus 5.5," 22 Sep 2026.